

SAM — FEATURE IMPLEMENTATION MATRIX

Technical Architecture & Verification Mapping

Target: SBTET / Polytechnic Institutions

Department: Computer Science and Engineering

Stack: React 19 • TypeScript • Vite 8 • Firebase • Cloudinary

Status: 184/184 Automated Tests Passed

SAM — FEATURE IMPLEMENTATION MATRIX

Verification Date: 2026-09-24

Source Base: C:\Users\ASUS\.gemini\antigravity-ide\scratch\sam-app\

Build Status: Passing (Vite 8 + React 19)

Automated Test Results: 184 / 184 Assertions Passed across 10 Suites

This document provides a technical classification of every feature in the Smart Assignment Manager (SAM) application. Every feature is categorized as one of:

- IMPLEMENTED** : Actively functional, confirmed in code, and verified by tests.
- LEGACY** : Maintained strictly for backward compatibility with pre-existing data.
- PARTIALLY IMPLEMENTED** : Functional core with additional enhancements planned.
- PLANNED / NOT IMPLEMENTED** : Architectural concept reserved for future releases.

1. COMPREHENSIVE FEATURE MATRIX

Feature Category	Specific Feature	Status	Technical Implementation & How It Works	Key Source Files
Authentication	Student Synthetic Email Login	IMPLEMENTED	Converts diploma PIN (e.g. <code>24170-CM-001</code>) into synthetic email <code>24170-cm-001@student.sam.edu.in</code> and authenticates via Firebase Auth <code>signInWithEmailAndPassword</code> .	authService.ts , StudentLoginPage.tsx
Authentication	Staff Synthetic Email Login	IMPLEMENTED	Converts 10-digit mobile number into synthetic email <code>staff_@sam.edu.in</code> and authenticates via Firebase Auth.	authService.ts , TeacherLoginPage.tsx
Authentication	Admin Synthetic Email Login	IMPLEMENTED	Converts 10-digit mobile number into synthetic email <code>admin_@sam.edu.in</code> and authenticates via Firebase Auth.	authService.ts , AdminLoginPage.tsx
Authentication	Authoritative Identity (UID)	IMPLEMENTED	Never derives authorization from client state or <code>localStorage</code> ; strictly binds security and Firestore queries to <code>request.auth.uid</code> .	AuthContext.tsx , firebaseConfig.ts
Authentication	Same-Device Isolation	IMPLEMENTED	<code>signOut</code> clears all React states, drilldowns, and caches. New login re-queries data strictly constrained by the new user's UID.	App.tsx , AuthContext.tsx
Admin Module	Staff Provisioning	IMPLEMENTED	Admin provisions faculty records with name, mobile, and department. Creates <code>/users</code> and <code>/staff</code> records.	academicService.ts , AdminStaffPage.tsx
Admin Module	Student Provisioning	IMPLEMENTED	Admin registers students with diploma PIN, name, semester, and mobile. Enforces valid PIN pattern (<code>24170-CM-XXX</code>).	academicService.ts , AdminStudentsPage.tsx
Admin Module	Class Management	IMPLEMENTED	Full CRUD on classrooms (1st, 3rd, 4th, 5th Semesters in CSE). Records student and assignment count statistics.	AdminClassesPage.tsx
Admin Module	Subject Management	IMPLEMENTED	Full CRUD on curricular subjects (name, code, semester, department). Toggle active/disabled status.	AdminSubjectsPage.tsx
Admin Module	Teaching Allocations	IMPLEMENTED	Admin allocates specific staff members to subject + semester + class cohorts in <code>/teachingAssignments</code> .	AdminTeachingAssignmentsPage.tsx
Admin Module	Global Assignment Overlook	IMPLEMENTED	Admin views, filters, audits, and deletes assignments across all institutional faculty.	AdminAssignmentsPage.tsx
Admin Module	Global Submission Audit	IMPLEMENTED	Admin inspects student submissions, high-resolution pages, verification codes, and evaluation timestamps.	AdminSubmissionsPage.tsx
Staff Module	Allocation Dashboard	IMPLEMENTED	Dashboard dynamically filters viewable classes and creation options to authorized teaching loads only.	TeacherDashboard.tsx

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Staff Module	Assignment Creation	IMPLEMENTED	Modal allows setting title, instructions, maxMarks (default 10), dueDate, and target subject/class.	CreateAssignmentModal
Staff Module	Question Image Upload	IMPLEMENTED	Allows staff to upload, preview, replace, or remove an optional question photo/diagram (JPG, PNG, WEBP, max 10MB) stored strictly under Cloudinary <code>sam/questions</code> .	CreateAssignmentModal cloudinary.ts, imageCom
Staff Module	Save Draft Mode	IMPLEMENTED	Stores assignment with <code>status = 'draft'</code> and <code>published = false</code> . Completely hidden from students.	assignmentService.ts
Staff Module	Publish Mode	IMPLEMENTED	Transitions assignment to <code>status = 'published'</code> , activates class targeting, and triggers notifications.	assignmentService.ts
Staff Module	Submissions Queue	IMPLEMENTED	Lists all student submissions filtered by assignment with pending review count badges.	TeacherSubmissionsPage
Student Module	Dashboard & 6 Statuses	IMPLEMENTED	Categorizes assignments across 6 distinct tabs: <code>ALL</code> , <code>PENDING</code> , <code>SUBMITTED</code> , <code>EVALUATED</code> , <code>RETURNED</code> , <code>CLOSED</code> .	StudentDashboard.tsx, AssignmentCard.tsx
Student Module	Exact Class Targeting	IMPLEMENTED	Queries assignments strictly by <code>student.semester</code> and <code>student.department</code> . Blocks cross-class visibility.	StudentDashboard.tsx, firebase.rules
Verification Code	6-Character Code Generator	IMPLEMENTED	Cryptographically generates 6 unambiguous alphanumeric characters (<code>23456789ABCDEFGHIJKLMNOPQRSTUVWXYZ</code>).	verificationCodeGenerat
Verification Code	Code Persistence	IMPLEMENTED	Bound to <code>(studentId, assignmentId)</code> in <code>/verification_codes</code> . Persists across app reload, retake, and resubmit.	verificationCodeService.t
Camera Submission	Rear Camera Preference	IMPLEMENTED	Uses <code>navigator.mediaDevices.getUserMedia</code> with <code>facingMode: { ideal: "environment" }</code> .	SubmitAssignmentModa
Camera Submission	No Gallery / File Picker	IMPLEMENTED	Prohibits Choose File No file chosen for new submissions. Guarantees live camera capture of physical pages.	SubmitAssignmentModa
Camera Submission	Multi-Page Capture Strip	IMPLEMENTED	Capture multiple pages, navigate page 1 of N, delete individual pages, retake, and review thumbnails.	SubmitAssignmentModa
Camera Submission	Client-Side Watermarking	IMPLEMENTED	<code>drawVerificationWatermark</code> draws Student Name, PIN, Verification Code, and Timestamp in bottom-right corner.	imageCompressor.ts
Camera Submission	Image Compression	IMPLEMENTED	Dynamic canvas scaling (max 1600px) and progressive JPEG quality (0.55-0.82) targeting ~150 KB with ink legibility.	imageCompressor.ts

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Cloud Storage	Cloudinary Direct Upload	IMPLEMENTED	Uploads multipart/form-data directly to Cloudinary REST API with unsigned preset. Returns <code>secure_url</code> .	cloudinary.ts
Legacy Submissions	Google Drive Compatibility	LEGACY	Pre-existing Drive submissions remain viewable; no new Drive link inputs exist in new submission modal.	submissionService.ts , NotebookImageViewer.ts
Teacher Evaluation	Full-Screen Workspace	IMPLEMENTED	100vw × 100vh digital evaluation workspace with top header, persistent <code>DONE</code> button, and queue navigation.	GradeSubmissionModal.ts
Digital Annotation	Transparent Canvas Overlay	IMPLEMENTED	Synchronized HTML5 canvas overlay positioned above student page. Zoom/pan moves image and annotations in sync.	AnnotationCanvas.tsx
Digital Annotation	Pen Tool & Colors	IMPLEMENTED	4 colors (Red <code>#EF4444</code> , Blue <code>#3B82F6</code> , Green <code>#10B981</code> , Black <code>#111827</code>) and 3 stroke widths (2px, 4px, 7px).	AnnotationCanvas.tsx
Digital Annotation	Vector Eraser	IMPLEMENTED	Erases only teacher annotation strokes without modifying underlying student image.	AnnotationCanvas.tsx
Digital Annotation	Text Tool & Presets	IMPLEMENTED	Click image to type feedback notes; includes quick preset feedback chips ("Check calculation", etc.).	AnnotationCanvas.tsx
Digital Annotation	Undo / Redo Stack	IMPLEMENTED	Multi-level undo/redo state stacks with keyboard shortcut listeners (<code>Ctrl+Z</code> , <code>Ctrl+Y</code>).	AnnotationCanvas.tsx
Digital Annotation	Non-Destructive Storage	IMPLEMENTED	Saves vector strokes in <code>/submissionAnnotations/{submissionId}</code> . Never overwrites original Cloudinary image.	annotationService.ts
Grading	Marks Input & Validation	IMPLEMENTED	Enforces <code>0 ≤ text{marks} ≤ text{maxMarks}</code> . Sets status to <code>'checked'</code> with teacher UID and timestamp.	submissionService.ts
Evaluation	Return for Correction	IMPLEMENTED	Sets status to <code>'returned'</code> with mandatory feedback. Prompts student to correct answers and resubmit.	submissionService.ts
Resubmission	Submission History	IMPLEMENTED	Retains all previous submission rounds, images, marks, and feedback in <code>history: SubmissionHistoryItem[]</code> .	submissionService.ts
Verification	AI Handwriting OCR Helper	IMPLEMENTED	Optional Tesseract.js OCR helper scans submitted page for handwritten verification code with fuzzy matching.	aiVerificationService.ts
Notifications	In-App Alerts	IMPLEMENTED	Dispatches notifications to <code>/notifications</code> on assignment publish, submission, grading, and return.	notificationService.ts
PWA	Installability & Manifest	IMPLEMENTED	Standalone display mode, high-res icons, theme color, custom install banners and launch toasts.	manifest.json , PWAInstall

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Security	Firestore Security Rules	IMPLEMENTED	Production security rules enforce role-based access, student data isolation, and marks write protection.	firestore.rules
Automated Testing	Regression & Flow Tests	IMPLEMENTED	10 Node.js test suites executing 184 total assertions verifying all critical flows with 100% pass rate.	package.json , <code>test_*.m</code>
Export	Bulk Assignment PDF Export	PLANNED	Institutional export of all evaluated papers as compiled PDF bundles for academic audits.	Architecture reserved
AI Assessment	Automated AI Grading Suggester	PLANNED	Multimodal LLM analysis to pre-read handwritten solutions and provide preliminary grading recommendations.	Architecture reserved